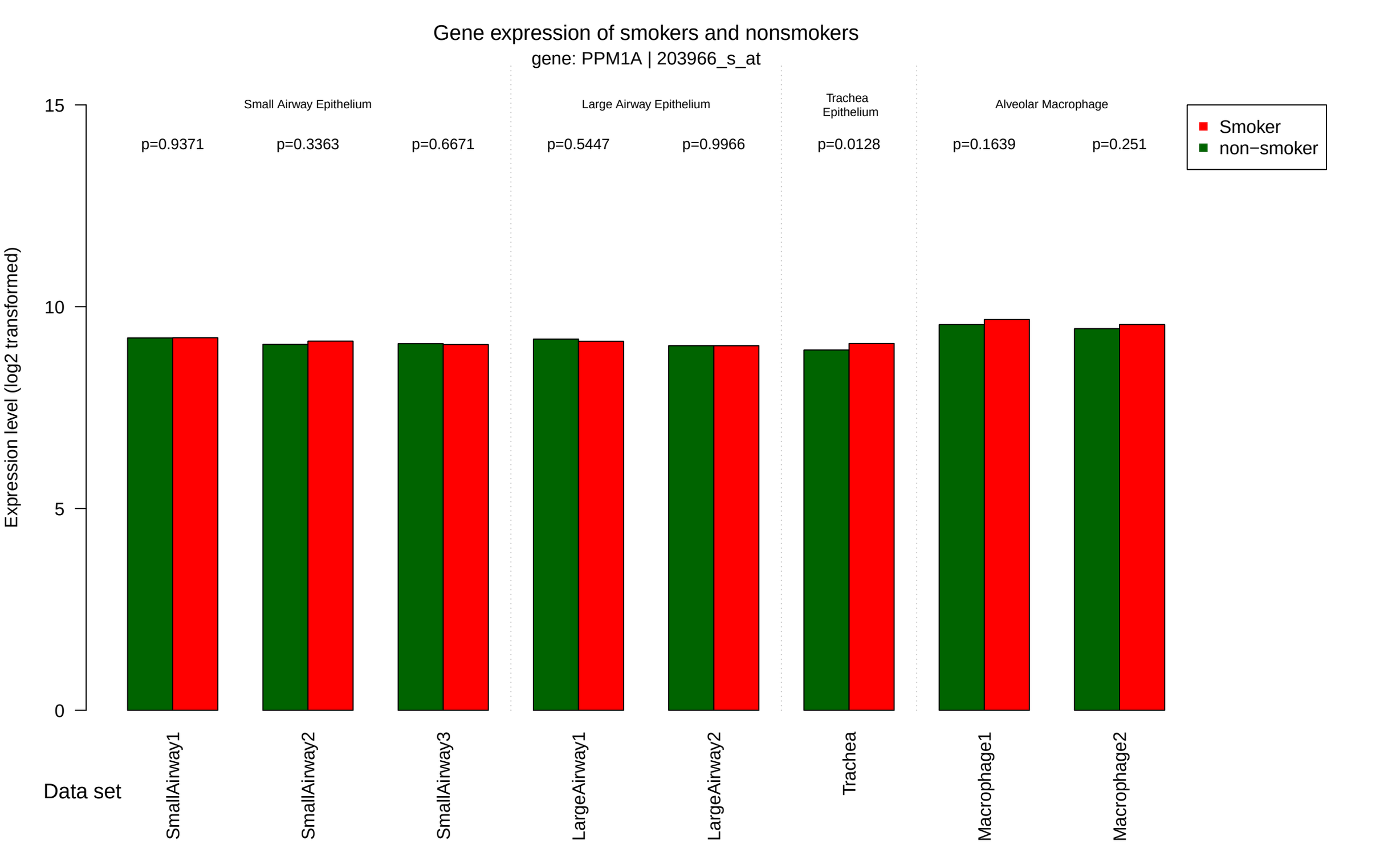
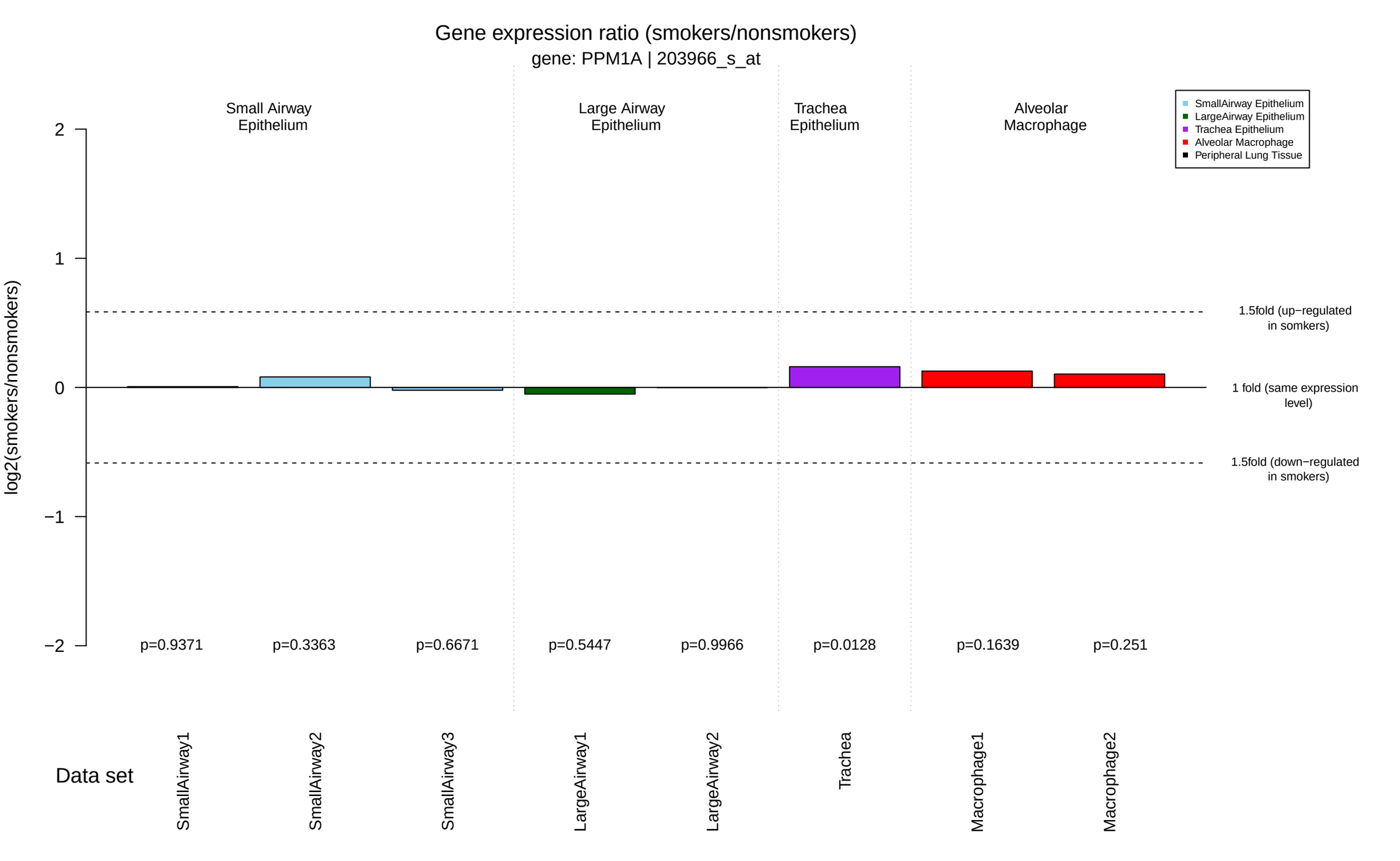
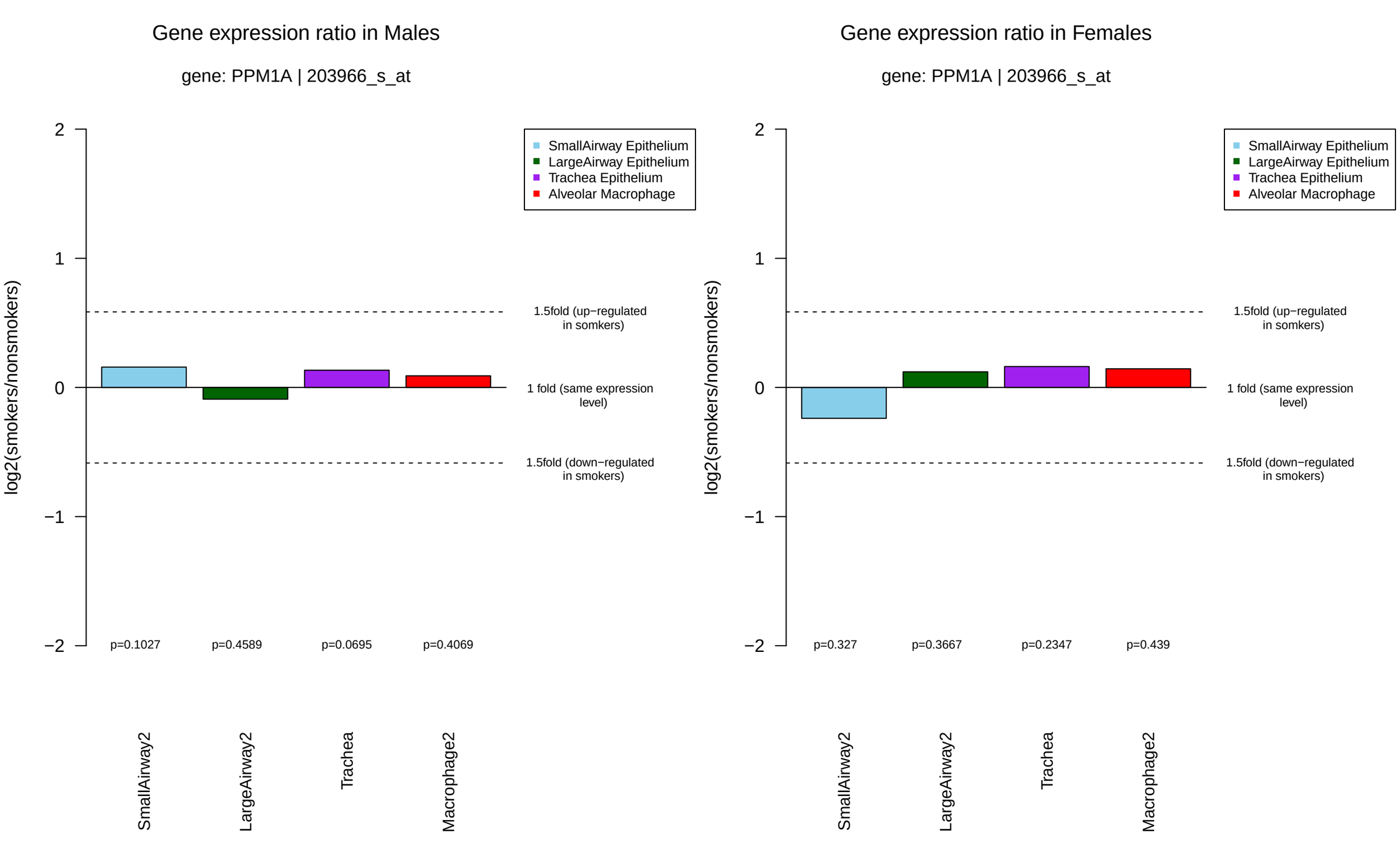
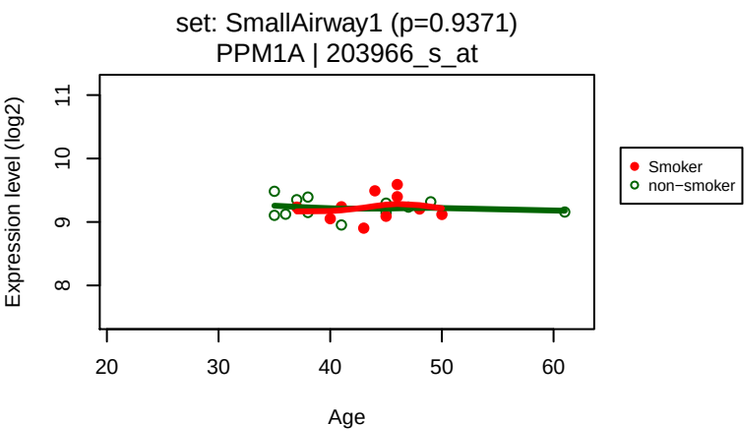
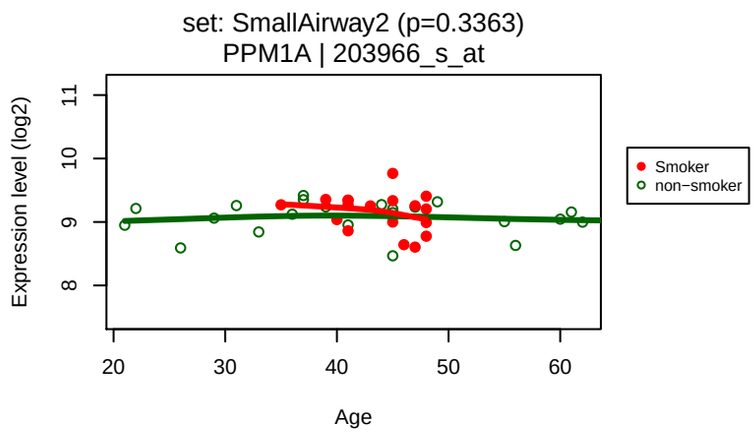
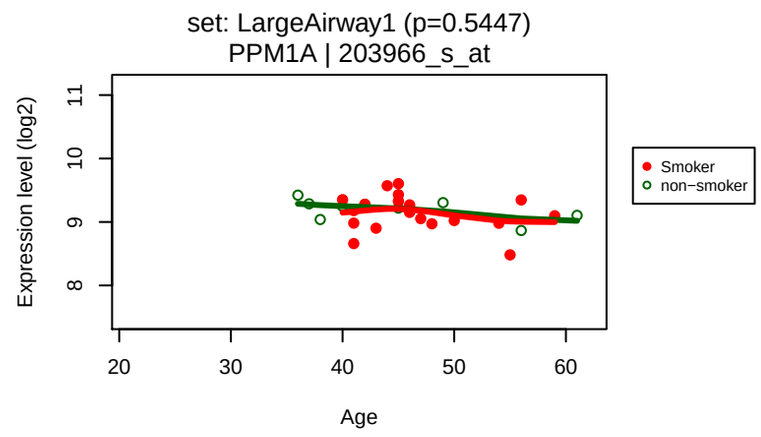

Gene expression of smokers and nonsmokers
gene: PPM1A | 203966_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9371
p=0.3363
p=0.6671
p=0.5447
p=0.9966
p=0.0128
p=0.1639
p=0.251
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

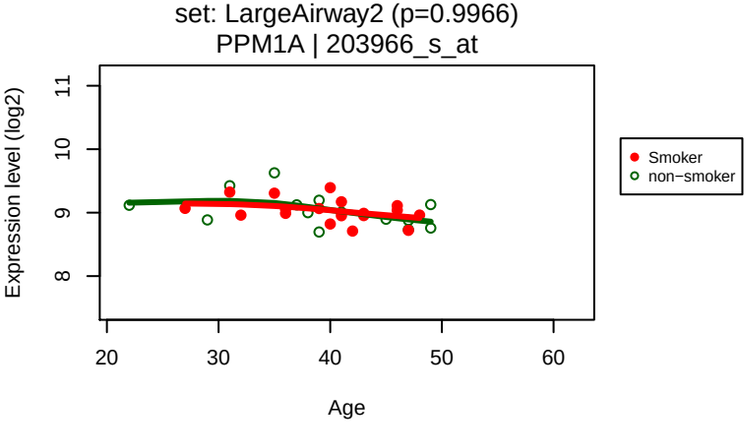

Gene expression ratio (smokers/nonsmokers)
gene: PPM1A | 203966_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9371
p=0.3363
p=0.6671
p=0.5447
p=0.9966
p=0.0128
p=0.1639
p=0.251
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

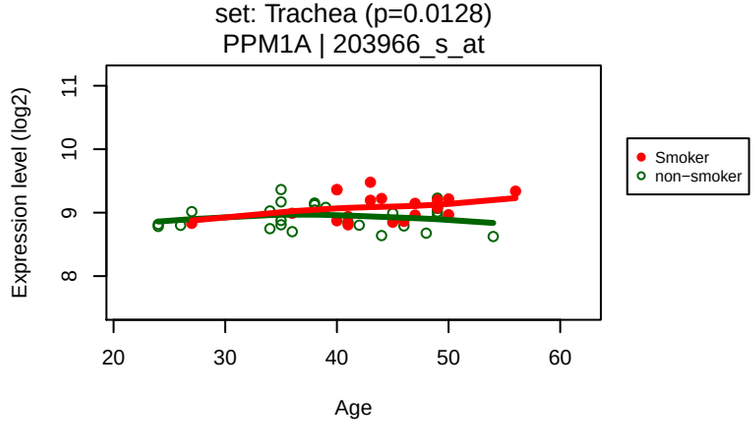

Gene expression ratio in Males
gene: PPM1A | 203966_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1027
p=0.4589
p=0.0695
p=0.4069
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1A | 203966_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.327
p=0.3667
p=0.2347
p=0.439
SmallAirway2
LargeAirway2
Trachea
Macrophage2

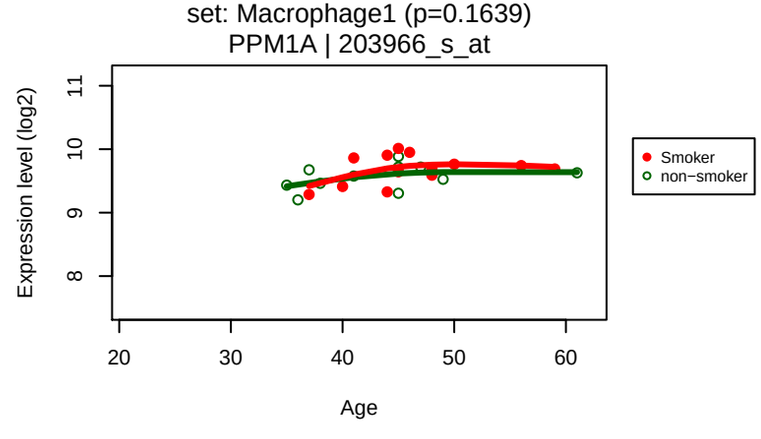

set: SmallAirway1 (p=0.9371)
PPM1A | 203966_s_at
Expression level (log2)
Age
Smoker
non−smoker

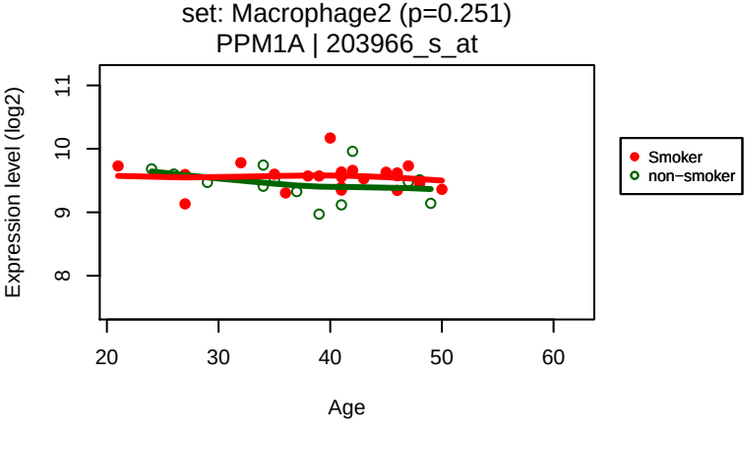

set: SmallAirway2 (p=0.3363)
PPM1A | 203966_s_at
Expression level (log2)
Age
Smoker
non−smoker

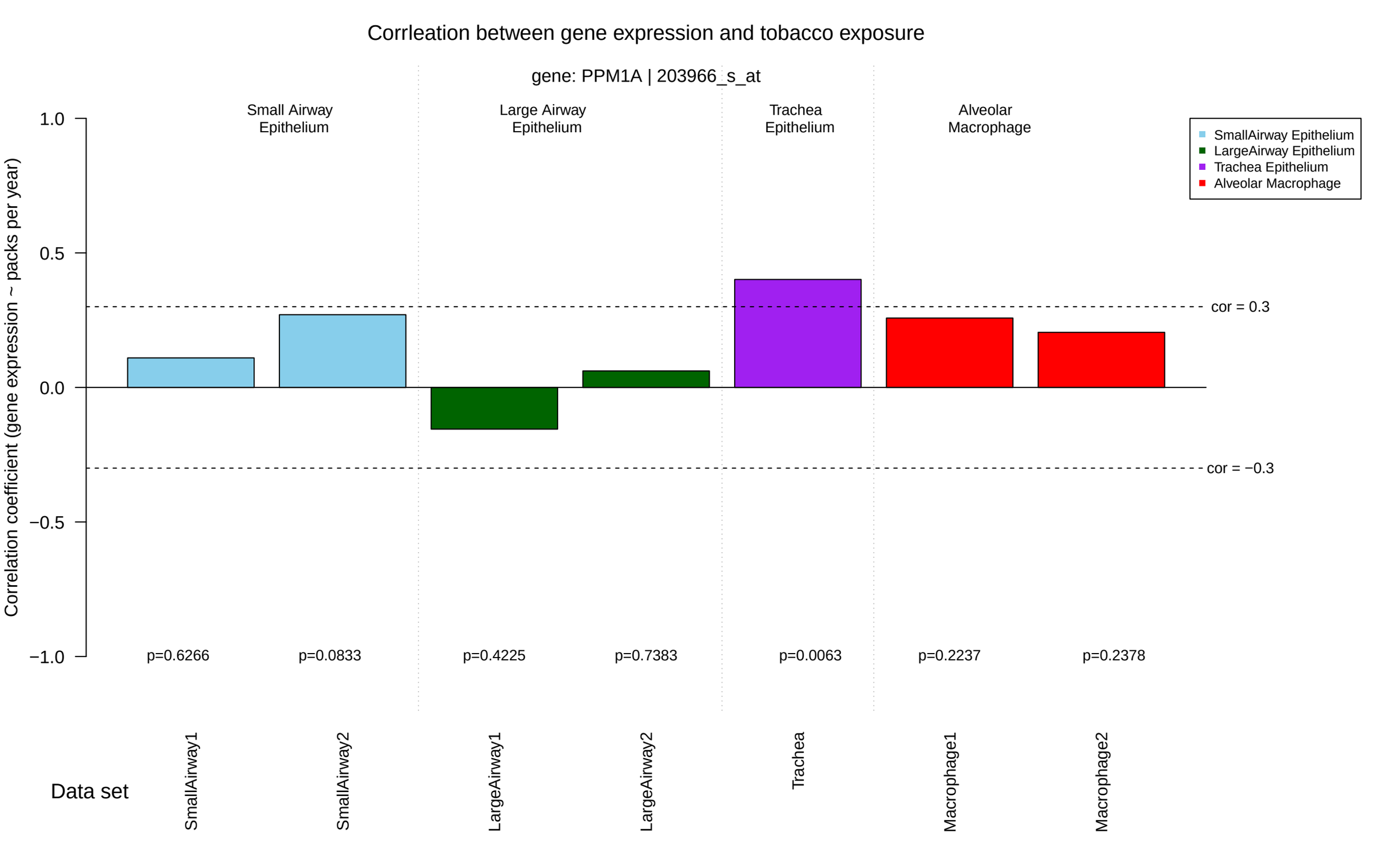

set: LargeAirway1 (p=0.5447)
PPM1A | 203966_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9966)
PPM1A | 203966_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0128)
PPM1A | 203966_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1639)
PPM1A | 203966_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.251)
PPM1A | 203966_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1A | 203966_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6266
p=0.0833
p=0.4225
p=0.7383
p=0.0063
p=0.2237
p=0.2378
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2